REPORT

AMAZON PRODUCT RATING ANALYSIS & PREDICTION

USING MACHINE LEARNING

Introduction to Artificial Intelligence & Machine Learning

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Semester : 2

Technologies Used:

Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-Learn

1. INTRODUCTION

E-commerce platforms such as Amazon generate massive volumes of structured data every day, including product pricing, discounts, categories, and customer ratings. This data contains valuable insights that can be used to understand consumer behavior and improve business decision-making.

The objective of this project is to analyze Amazon product data using data analysis and machine learning techniques. The project focuses on identifying patterns in product pricing, discounts, and ratings, and building a predictive model to classify whether a product is likely to receive a high rating.

This project demonstrates the application of Artificial Intelligence and Machine Learning concepts learned in the academic curriculum to solve real-world data problems.

2. PROBLEM STATEMENT

The main problem addressed in this project is to analyze how product-related attributes such as price, discount, and category influence customer ratings.

Additionally, the project aims to develop a machine learning model capable of predicting whether a product will receive a high rating (≥ 4.0) based on its features.

This helps in understanding:

- Whether discounts influence ratings
- Whether price impacts customer satisfaction
- Which features are most important in determining product quality perception

3. DATASET DESCRIPTION

The dataset used in this project is obtained from Kaggle and contains approximately 1,465 records of Amazon products.

Each record represents a product with attributes related to pricing, discounts, and customer feedback.

```
[48]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   product_id           1465 non-null   object
 1   product_name         1465 non-null   object
 2   category             1465 non-null   object
 3   discounted_price     1465 non-null   object
 4   actual_price         1465 non-null   object
 5   discount_percentage  1465 non-null   object
 6   rating               1465 non-null   object
 7   rating_count         1463 non-null   object
 8   about_product        1465 non-null   object
 9   user_id              1465 non-null   object
 10  user_name            1465 non-null   object
 11  review_id            1465 non-null   object
 12  review_title         1465 non-null   object
 13  review_content       1465 non-null   object
 14  img_link             1465 non-null   object
 15  product_link         1465 non-null   object
dtypes: object(16)
memory usage: 183.3+ KB
```

3.1 Domain of Dataset

The dataset belongs to the **E-commerce and Retail domain**, focusing on product listings, pricing strategies, and customer rating behavior.

3.2 Dataset Features

The dataset includes the following attributes:

- Product ID
- Product Name
- Category
- Actual Price
- Discounted Price
- Discount Percentage
- Rating
- Rating Count

3.3 Data Characteristics

The dataset consists of both numerical and categorical variables. Some missing values were observed in certain attributes, especially discount-related fields, which required preprocessing.

4. DATA COLLECTION AND PREPROCESSING

Data preprocessing is a crucial step in machine learning to ensure data quality and consistency.

4.1 Handling Missing Values

Missing values present in the dataset were handled using appropriate techniques:

- Numerical values were replaced using mean imputation
- Categorical values were replaced using mode imputation
- In some cases, incomplete records were removed

4.2 Data Cleaning

The dataset contained special characters such as:

- ₹ in price columns
- % in discount percentage

These were removed, and values were converted into numerical format for analysis.

4.3 Data Transformation

To make the dataset suitable for machine learning:

- Text values were encoded or cleaned
- Numerical columns were standardized

- New derived features were created for better prediction

```
df["discounted_price"] = (df["discounted_price"].str.replace("₹","").str.replace(",","").astype(float))

df["actual_price"] = (df["actual_price"].str.replace("₹","").str.replace(",","").astype(float))

df["discount_percentage"] = (df["discount_percentage"].str.replace("%","").astype(float))

df["rating"] = pd.to_numeric(df["rating"],errors="coerce")
```

5. STATISTICAL ANALYSIS OF DATASET

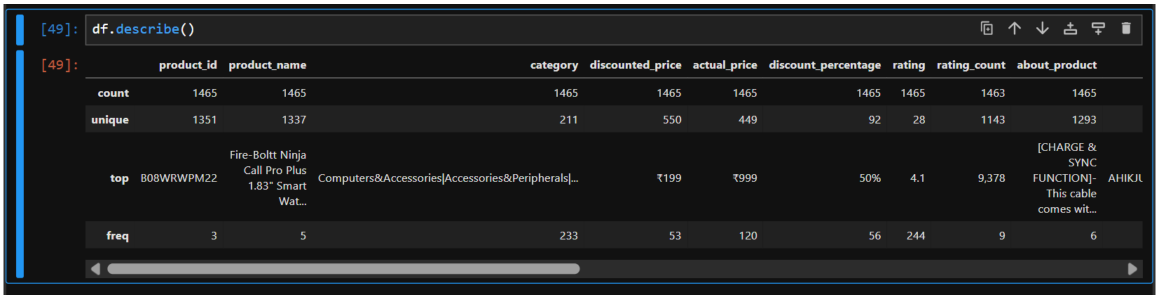
A statistical analysis was performed to understand the distribution and behavior of the dataset.

The following statistical measures were computed:

- Count
- Mean
- Median
- Mode
- Minimum and Maximum values
- Range
- Variance
- Standard deviation

Observations:

- Product prices show high variation across categories
- Ratings are mostly concentrated between 4 and 5
- Discount percentages vary significantly across products



[49]:
```
df.describe()
```

[49]:

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product	
count	1465	1465	1465	1465	1465	1465	1465	1463	1465	
unique	1351	1337	211	550	449	92	28	1143	1293	
top	B08WRWPM22	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹999	50%	4.1	9,378	[CHARGE & SYNC FUNCTION]- This cable comes wit...	AHIKJL
freq	3	5	233	53	120	56	244	9	6	

6. EXPLORATORY DATA ANALYSIS (EDA)

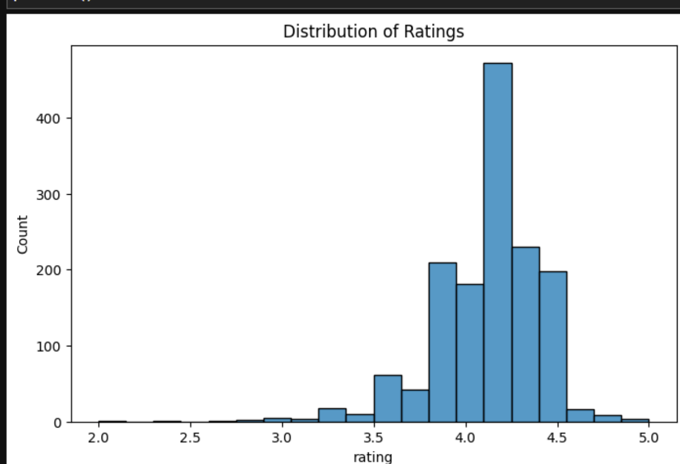
Exploratory Data Analysis was performed to extract meaningful insights using visualization techniques.

6.1 Rating Distribution

Most products have high ratings, indicating generally positive customer feedback.

Section 4: Exploratory Data Analysis

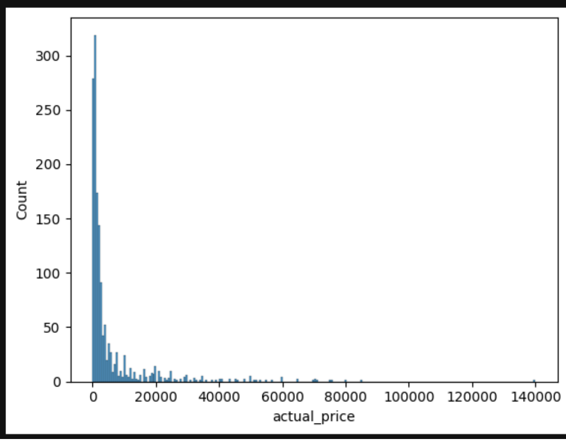
[87]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Rating Distribution

plt.figure(figsize=(8,5))
sns.histplot(df["rating"], bins=20)
plt.title("Distribution of Ratings")
plt.show()
```



6.2 Price Distribution

Product prices are widely distributed, with most products in the lower and mid-price range.

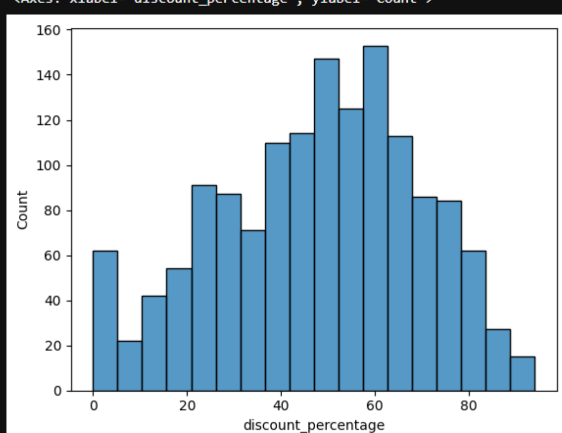
```
[88]: # Actual Price Distribution
      sns.histplot(df["actual_price"])

[88]: <Axes: xlabel='actual_price', ylabel='Count'>
```



6.3 Discount Analysis

Discounts are commonly applied across products, with varying intensity.

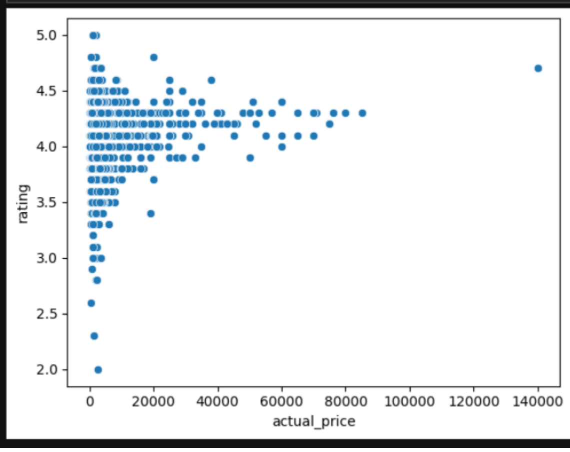
```
[89]: # Discount Distribution
      sns.histplot(df["discount_percentage"])

[89]: <Axes: xlabel='discount_percentage', ylabel='Count'>
```



6.4 Price vs Rating

There is no strong linear relationship between price and rating.

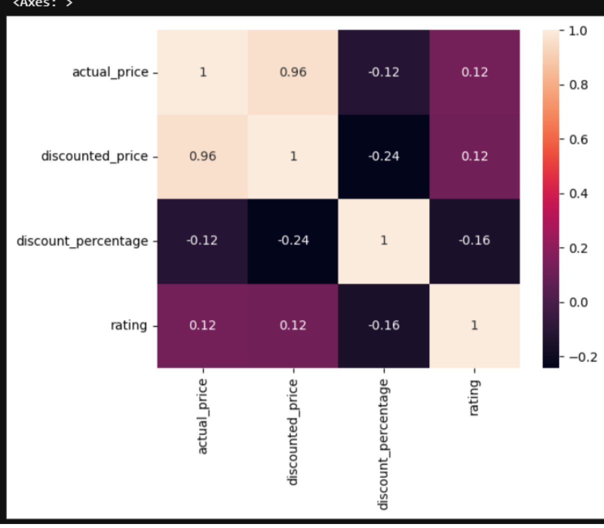
```
[90]: # Price vs Rating
      sns.scatterplot(x="actual_price",y="rating",data=df)
      plt.show()
```



6.5 Correlation Analysis

A heatmap was used to analyze relationships between numerical variables. Weak correlation was observed between price-related features and ratings.

```
[92]: # Correlation Heatmap

      numeric_cols = ["actual_price","discounted_price","discount_percentage","rating"]

      sns.heatmap(df[numeric_cols].corr(),annot=True)

[92]: <Axes: >
```



7. FEATURE ENGINEERING

Feature engineering was performed to improve model performance and extract meaningful patterns.

New features created include:

- Savings = Actual Price − Discounted Price
- Price Ratio = Discounted Price / Actual Price
- High Rating Label (Target Variable)

The target variable was defined as:

- 1 → Rating ≥ 4 (High Rating)
- 0 → Rating < 4 (Low Rating)

These transformations helped convert the problem into a binary classification task.

8. MACHINE LEARNING MODEL IMPLEMENTATION

Multiple supervised machine learning models were implemented to predict high-rated products.

8.1 Models Used

The following classification models were trained:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Classifier
- Random Forest Classifier

8.2 Training Process

The dataset was split into training and testing sets in an 80:20 ratio. Feature scaling was applied to normalize numerical values.

9. MODEL EVALUATION AND PERFORMANCE METRICS

The models were evaluated using various performance metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score

9.1 Model Comparison

Random Forest performed better compared to other models due to its ability to handle feature interactions effectively.

[83]:

	Model	Accuracy
3	Random Forest	0.754266
0	Logistic Regression	0.747440
1	KNN	0.720137
2	Decision Tree	0.706485

10. RESULTS AND DISCUSSION

The analysis shows that pricing factors alone are not sufficient to determine product ratings. Customer satisfaction depends on multiple factors.

Key findings:

- Most products have high ratings (above 4)
- Discounts do not guarantee better ratings
- Machine learning models can moderately predict rating category
- Random Forest performed best among all models

11. CONCLUSION

This project successfully demonstrates the application of Artificial Intelligence and Machine Learning techniques on real-world e-commerce data.

The study concludes that:

- Data preprocessing plays a critical role in model performance
- Feature engineering improves prediction accuracy
- Ensemble models perform better than basic classifiers

Overall, the project provides valuable insights into Amazon product data and demonstrates practical implementation of ML concepts.

12. FUTURE SCOPE

The project can be further improved in the following ways:

- Applying Natural Language Processing (NLP) on product reviews
- Building recommendation systems
- Deploying the model using Streamlit or Flask
- Using larger and more diverse datasets
- Integrating real-time prediction systems

13. REFERENCES

- Kaggle Dataset Repository
- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib & Seaborn Documentation
- Artificial Intelligence & Machine Learning Course Material